

Beyond Spatial Accuracy: Diagnosing Persistent Orientation Failures in Vision–Language Models

Anonymous WACV Datasets Track submission

Paper ID *****

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Supplementary Material

001	1. Exact Prompts, Decoding, and Parsing	045
002	VSR plain prompt (identical for 2B and 7B, all condi-	046
003	tions; '{statement}' is the VSR caption):	047
004	Look at the image carefully.	048
005		049
006	Statement: "{statement}"	050
007		051
008	Is this statement true or false?	052
009		053
010	Answer with exactly one word:	054
011	True or False.	055
012	2B structured-decomposition prompt (2B only; sin-	056
013	gle run):	057
014	Look carefully at the image.	
015		
016	Statement:	
017	"{statement}"	
018		
019	Determine whether the statement is true	
020	by doing the following:	
021	1. Identify the subject object.	
022	2. Identify the reference object.	
023	3. Determine the position and	
024	orientation of the subject relative	
025	to the reference object.	
026	4. Pay careful attention to left/right,	
027	front/back, viewpoint, and	
028	orientation.	
029	5. Decide whether the stated spatial	
030	relationship matches the image.	
031		
032	Answer with exactly one word:	
033	True or False.	
034	SITE multiple-choice prompt (replicated from the	
035	official lmms-eval task sitebench/utlis.py):	
036	# image examples (one <image> token per visual):	
037	<image>...	
038	Question: {question}	
039	Options:	
040	A: {opt0}	
041	B: {opt1}	
042	...	
043	Give me the answer letter directly.	
044	The best answer is:	
	# video examples (pre-prompt):	
	Select the best answer to the following	
	multiple-choice question based on the	
	video. Respond with only the letter	
	of the correct option.	
	Question: {question}	
	Options:	
	A: ...	
	B: ...	
	...	
	Give me the answer letter directly.	
	The best answer is:	
	Decoding. All generations are greedy:	
	do_sample=False, temperature 0, top-p 1.0.	
	max_new_tokens: 5 for the 2B wrapper, 128 for	
	7B VSR runs, 16 for SITE (change validated parse-	
	identical on 125/125 held-out outputs before adoption;	
	recorded in the SITE run metadata).	
	MiMo-V2.5 protocol (external-validation row).	
	MiMo-V2.5 (XiaomiMiMo/MiMo-V2.5, 310B total	
	/ 15B activated, dedicated 729M vision encoder) is	
	evaluated zero-shot through its public API with the	
	identical frozen VSR prompt above, thinking mode dis-	
	abled (chat_template_kwargs.enable_thinking=false	
	and thinking.type=disabled; verified content-only,	
	zero reasoning tokens, deterministic on re-runs),	
	temperature 0, top-p 1.0, max_tokens=64, one image	
	per request (canonical COCO URLs, with automatic	
	base64 retry fallback from the local cache), and 12	
	parallel workers (within the provider's rate limits).	
	Outputs use the same parser; 3.1% of responses	
	did not conform to the one-word format and are	
	counted as wrong (never guessed), and 294 responses	
	contained chain-of-thought text alongside a parseable	
	verdict; both are flagged in the committed predictions	
	(results/mimo/). The evaluation cost is reported in	
	results/mimo/usage_report.json (per-request token	
	accounting).	
	MiMo-V2.5 output-format sensitivity (addi-	
	tive). Because 69 of the 2,195 outputs (3.1%)	
	were non-conforming, we report the parseable-	
	only counterpart of the headline numbers	
	(results/mimo/mimo_parseable_sensitivity.json):	
	overall accuracy rises from 79.5% (1746/2195, invalid	
	counted wrong) to 82.1% (1746/2126), and orientation	
	is unchanged at 65.7% (90/137) because none of the 69	

non-conforming outputs fell in the orientation family. The orientation conclusions are therefore independent of the format-policy choice; the overall row depends on it by 2.6 pp.

Parsing. VSR outputs are parsed with an anchored case-insensitive regex (src/evaluation/parser.py): strips prefixes ('the answer is', 'answer:', 'response:', 'output:') and trailing punctuation; accepts exact 'true/yes/correct' or 'false/no/incorrect' forms; substring fallback only when exactly one of 'true'/'false' appears. Invalid outputs are counted as wrong, never guessed; invalid rate is 0 on all reported runs. SITE outputs use the official parse_multi_choice_response (letter); unparseable responses are counted incorrect and reported.

2. Training and Adapter Configurations

All LoRA conditions share: $r=8$, $\alpha=16$, dropout 0.05, AdamW lr 10^{-4} , weight decay 0.01, linear warmup 10% of steps, 2 epochs, bf16, gradient checkpointing, gradient clipping 1.0, vision tower frozen. Adapter configs are committed at checkpoints/*/final/adapter_config.json.

Manifest construction. General: 2,000 statements sampled from the VSR train split. Targeted: orientation-over-sampled subset of the same split. Hard-negative: the orientation block of the general manifest is replaced by audited-clean originals plus paired hard negatives (facing $\leftrightarrow$ facing away from; parallel $\leftrightarrow$ perpendicular) with flipped labels; every included example was audited for label correctness (451 train orientation examples audited: 402 clean, 28 excluded, 21 original-only). 2B runs split 5% of the manifest for validation (seed 42). All manifests derive exclusively from the VSR train split; all evaluations use test only.

3. Probe Methodology and Full Tables

Features. Qwen2-VL-7B-Instruct, vision tower frozen. ViT patch embeddings (1280-d) and post-merger features (3584-d), mean-pooled per image (ungrounded) or per grounding box (grounded). Grounding boxes from frozen Qwen2-VL grounding (94.6% of 647 orientation images yield both boxes; failures are mostly genuinely absent objects), rescaled to pixels; region features mean-pooled over patch centers inside each box.

Tasks. T1 facing vs. facing away (test $n=103$, majority 64.2%); T2 parallel vs. perpendicular ($n=34$, majority 64.7%); T3 4-way ($n=137$, majority 49.6%). Training on audited-clean VSR train; 5-fold CV; no test examples in training.

Readouts. Linear logistic regression; MLP (256

hidden, ReLU, Adam, early stopping on CV); patch-vote (per-patch linear probe, majority vote over patches); grounded variants with feature sets visual ([subj, ref, subj-ref, subj-ref]), geometry (12-d normalized box features), and visual+geometry.

Ungrounded results (test acc %; balanced acc in parentheses; Wilson 95% CI; majority baselines: T1 64.2%, T2 64.7%, T3 49.6%):

T1 facing vs. away ($n=103$)	Test	Bal. acc	95% CI
ViT linear	65.0	60.4	[55, 74]
ViT MLP	55.3	52.0	[45, 65]
Merger linear	68.9	64.5	[59, 77]
Merger MLP	52.4	49.2	[42, 62]
ViT patch-vote	62.1	—	[53, 71]
Merger patch-vote	60.2	—	[51, 69]

T2 parallel vs. perpendicular ($n=34$)	Test	Bal. acc	95% CI
ViT linear	61.8	61.0	[44, 77]
ViT MLP	67.6	69.3	[50, 82]
Merger linear	64.7	61.4	[47, 79]
Merger MLP	64.7	63.3	[47, 79]
ViT patch-vote	73.5	—	[57, 85]
Merger patch-vote	67.6	—	[51, 81]

T3 4-way ($n=137$)	Test	Bal. acc	95% CI
ViT linear	52.6	40.5	[44, 61]
ViT MLP	52.6	42.9	[44, 61]
Merger linear	53.3	44.0	[45, 61]
Merger MLP	47.4	43.2	[39, 56]
ViT patch-vote	51.8	—	[44, 60]
Merger patch-vote	52.6	—	[44, 61]

Grounded results (CV acc / test acc; majority T1 63.7%, T2 57.0%; visual = [subj, ref, subj-ref, subj-ref]; geometry = 12-d box features; T3 4-way is at chance for every readout, test acc 41.7–52.8% across all 12 cells vs. 49.9% majority, committed JSON):

Readout	T1 visual	T1 geom.	T2 visual	T2 geom.
ViT linear	59.8/61.6	61.1/60.6	51.2/53.6	44.2/50.0
ViT MLP	59.5/66.7	61.8/62.6	45.4/46.4	45.4/46.4
Merger linear	57.2/65.7	61.1/60.6	51.2/53.6	44.2/50.0
Merger MLP	56.3/52.5	61.1/71.7	49.0/53.6	50.0/42.9

The single above-majority cell (geometry-only merger MLP on T1, 71.7% test) has CV 61.1% below the 63.7% majority — not robust — and derives from the grounding model's own box statistics, not vision-feature content. In short, object-box conditioning with the model's own grounding predictions and mean-pooled region features does not recover robust orientation decodability under the tested readouts. The T2 patch-vote result (73.5%) is a suggestive point estimate with a wide CI overlapping the majority baseline ($n=34$).

Condition	LoRA targets	Trainable	Batch	Manifest
2B General	LM q/k/v/o	~2.3M	microbatch 4×4	general (2,000)
2B Targeted	LM q/k/v/o	~2.3M	same	targeted (2,000)
7B LM-only	LM q/k/v/o	~0.3M	1	general (2,000)
7B HardNeg	LM q/k/v/o	~0.3M	1	hard-negative (2,000)
7B Projector	visual.merger.mlp.0/2	152K	1	general (2,000)
7B Vision+Proj	merger + blocks 24–31	1.46M	1	general (2,000)

Table 1. LoRA conditions and their target modules.

Clear-subset control (canonical frozen snapshot). The 48-example automated (LLM-generated) audit identified ambiguous or questionable orientation labels; its reliability is assessed against two independent human re-audits in §6. Re-evaluation on progressively stricter subsets (values % from results/tables/orientation_clean_label_table.md, generated by scripts/clean_label_orientation.py): Because auditing changes the evaluated sample, strict-subset values are not a matched comparison against other relation families. The earlier probe-side clear/strict check (pre-freeze) additionally suggested that removing audit-flagged examples does not change the probe conclusion: the best ungrounded probe moved only 68.9% → 71.1% → 71.6% (full/clear/strict, $n = 103/97/88$) and the generative T1 readouts 64.1/66.0/68.2 (zero-shot) and 68.9/72.2/78.4 (General LoRA); committed in results/probe/clear_subset_results.json. The reliability of the automated audit underlying this table is assessed in §6 (human re-audits + inter-source agreement); the frozen table above is retained unchanged.

Human re-audit tables (versioned, additive). The exclusion masks of Table 2 were originally produced by an automated (LLM-generated) audit whose image grounding is unverified. Two independent human annotators re-rated all 137 orientation examples (binary clean/ambiguous flag) and the 48 persistent failures (eight-class taxonomy) from the images alone under a blind protocol. Applying the same taxonomy-derived strict mask to each human’s labels reproduces the qualitative conclusions, with best scores close to the automated one:

Under the human strict masks, orientation accuracy rises (best 77.6% for human A, 78.8% for human B, both 2B Targeted LoRA), error is not eliminated (the best human-mask score 78.8%, $n=104$, is on par with the automated 78.5%, $n=107$), and 2B structured still shows no gain (51.4% and 51.0%), so the paper’s sensitivity conclusions are not an artifact of the automated audit. The human binary masks alone do not reproduce the gains (best 66.7% human A / 66.2% human B vs. the 66.4% full-set best), which we read as fur-

ther evidence that clean-label judgments are labeler-dependent at the per-item level; per-item agreement and a consensus analysis are in §6.

4. Complete Statistical Test Battery

All pairwise tests are exact McNemar tests (binomial test on discordant pairs, two-sided). Discordant counts are reported as (condition-losses / condition-gains) relative to the control named in each row. No test was run and withheld; the battery below is complete for this paper.

Principal vs. exploratory comparisons. The battery above is fully enumerated; no test was run and withheld. The principal (confirmatory) comparisons are: zero-shot vs. LM-only consistency (ID 19) and the SITE transfer tests (§E). All other comparisons — vision-side conditions and their per-relation cells, hard-negative training vs. control, the remaining consistency comparisons, and the two-stage agreement rows (IDs 15–18) — are exploratory diagnostics reported with exact unadjusted p -values. Under a Holm adjustment over the 18 accuracy-valid VSR tests ($\alpha=0.05$), only the zero-shot-vs-LM-only consistency test survives (Holm $p=0.0018$, using the conservative $p<0.0001$ bound); the vision-side overall effects (raw $p=0.0043$, $p=0.0122$; Holm 0.073, 0.195) and all other comparisons do not survive a blanket adjustment and are not load-bearing. For the 3 SITE transfer tests, Holm-adjusted p -values are 0.012 (primary), 1.0, 1.0.

Single-checkpoint note. Every trained condition is a single checkpoint. All McNemar tests above quantify paired disagreement on the fixed test examples between two conditions; they do not estimate training-run variability, and the single-seed results reported here do not estimate training-run variability. Adapter-rank claims (e.g., the 1–2pp overall differences among trained 7B conditions) should be read in that light. A multi-seed extension (two additional seeds for 7B General LoRA and 7B Hard-Neg LoRA) is prepared in scripts/run_seed_variance.py and scripts/aggregate_seed_variance.py and requires a ≥ 40 GB GPU; its results, when available, will be reported additively.

Table 2. Clean-label orientation robustness (canonical frozen snapshot; values % from re-sults/tables/orientation_clean_label_table.md).

Condition	Full (137)	−Questionable (132)	Clear (124)	Strict (107)
2B zero-shot	62.8	65.2	66.9	75.7
2B structured	53.3	52.3	51.6	53.3
2B General LoRA	62.0	63.6	64.5	72.9
2B Targeted LoRA	64.2	65.9	68.5	76.6
7B zero-shot	63.5	65.2	68.5	72.0
7B General LoRA	65.7	67.4	70.2	78.5
7B Targeted LoRA	64.2	65.9	69.4	72.0
7B Hard-Neg LoRA	66.4	68.2	70.2	76.6
7B Projector LoRA	64.2	65.9	69.4	73.8
7B Vision+Projector LoRA	64.2	65.2	68.5	73.8
MiMo-V2.5 zero-shot	65.7	67.4	69.3	71.0

Table 3. Human re-audit clean-label orientation robustness (versioned, additive; values % from re-sults/tables/orientation_clean_label_table_human.md for human A and ..._human2.md for human B). The frozen automated-audit table (Table 2) is not modified.

Condition	Full (137)	−Q	Clear	Strict (107)	Human binary (75)
2B zero-shot	62.8	67.5	67.5	76.6	65.3
2B structured	53.3	53.2	53.2	51.4	66.7
2B General LoRA	62.0	66.7	66.7	74.8	66.7
2B Targeted LoRA	64.2	69.0	69.0	77.6	65.3
7B zero-shot	63.5	65.9	65.9	74.8	54.7
7B General LoRA	65.7	68.2	68.2	72.9	66.7
7B Targeted LoRA	64.2	65.1	65.1	69.2	57.3
7B Hard-Neg LoRA	66.4	68.2	68.2	72.9	65.3
7B Projector LoRA	64.2	65.1	65.1	71.0	61.3
7B Vision+Projector LoRA	64.2	67.5	67.5	72.9	64.0
MiMo-V2.5 zero-shot	65.7	68.2	68.2	72.0	58.7

Bootstrap family-relative contrasts (additive). The headline family-relative claim is quantified with a paired bootstrap over the 2,195 shared test examples (10,000 resamples, seed 20260812; re-sults/bootstrap_family_contrast.json, generated by scripts/bootstrap_family_contrast.py). For each family f , Δ_f is the accuracy change between two conditions and the contrast is $\Delta_f - \Delta_{\text{orientation}}$:

The only contrast whose 95% CI excludes zero is the scaling horizontal family (+13.9 pp, positive in 99.3% of resamples); depth and containment are directionally consistent but not individually significant, and under adaptation no family contrast is distinguishable from orientation. The family-relative claim is therefore carried by the horizontal family, and the paper reads the depth/containment differences as weaker supporting evidence rather than independently established.

Consistency verdict patterns. For each strict family and condition, the committed verdict table (results/consistency_verdict_table.json, scripts/consistency_verdict_table.py) reports the observed both-True / both-False / complementary counts, the original and flipped True rates, and the

verdict rates expected under verdict independence given those base rates:

The zero-shot pattern is dominated by a False-defaulting response bias: expected consistency under verdict independence is 30.2%, below the observed 36.9%, so a simple base-rate null does not explain the facing inconsistency; conversely, training raises consistency while introducing both-True contradictions (0% $\rightarrow$ 24.3% LM-only / 17.5% hard-neg), a disclosed specialization trade-off. The large sparse-MoE zero-shot model shows the same False-defaulting bias (0 both-True; 56.0% both-False; observed complementary 44.0% vs. 35.6% expected under independence), so scale alone does not alter the coherence failure mode. Full table for left/right ($n=245$), front/behind ($n=314$), and parallel/perp (soft, $n=34$) in the committed JSON.

Consistency definitions. Strict complements (exactly one truth value): left $\leftrightarrow$ right ($n=245$), in front of $\leftrightarrow$ behind ($n=314$), facing $\leftrightarrow$ facing away ($n=103$). Soft complement (parallel $\leftrightarrow$ perpendicular, $n=34$): both-False is legitimate; only both-True is a contradiction; both-True rates are reported separately. Self-

Table 4. Human B (second blind re-audit) clean-label orientation robustness (versioned, additive; values % from results/tables/orientation_clean_label_table_human2.md).

Condition	Full (137)	−Q	Clear	Strict (104)	Human binary (77)
2B zero-shot	62.8	66.4	67.2	77.9	62.3
2B structured	53.3	53.1	52.8	51.0	59.7
2B General LoRA	62.0	65.6	66.4	75.0	63.6
2B Targeted LoRA	64.2	68.8	68.8	78.8	63.6
7B zero-shot	63.5	65.6	66.4	76.0	58.4
7B General LoRA	65.7	68.0	68.8	74.0	62.3
7B Targeted LoRA	64.2	65.6	66.4	71.2	61.0
7B Hard-Neg LoRA	66.4	68.0	69.6	74.0	61.0
7B Projector LoRA	64.2	65.6	64.8	72.1	66.2
7B Vision+Projector LoRA	64.2	67.2	68.0	74.0	63.6
MiMo-V2.5 zero-shot	65.7	68.0	67.2	73.1	62.3

ID	Comparison	Subset	Discord.	<i>p</i>
1	HardNeg vs. Gen.	overall (2195)	52/60	0.508
2	HardNeg vs. Gen.	weak families	20/26	0.461
3	Projector vs. ctrl.	overall	—	0.0043
4	V+Proj vs. ctrl.	overall	—	0.0122
5	Projector vs. ctrl.	orientation	—	0.86
6	V+Proj vs. ctrl.	orientation	—	0.85
7–14	Proj./V+Proj vs. ctrl.	per-relation (4×2)	—	0.25–1.00
15	Two-stage A-lin. vs. ctrl.	agreement (127 boxed)	45/16	0.0003
16	Two-stage A-MLP vs. ctrl.	agreement (127 boxed)	47/14	<0.0001
17	Two-stage B-lin. vs. ctrl.	agreement (127 boxed)	44/20	0.0037
18	Two-stage B-MLP vs. ctrl.	agreement (127 boxed)	47/18	0.0004
19	Zero-shot vs. LM-only	consistency 662	117/43	<0.0001
20	HardNeg vs. LM-only	consistency 662	31/41	0.29
21	Projector vs. LM-only	consistency 662	63/48	0.18
22	V+Proj vs. LM-only	consistency 662	65/56	0.47

Table 5. Complete VSR test battery (22 rows). Per-relation cells (IDs 7–14) are enumerated in the committed vision_side_comparison.json: projector facing $p=0.77$, facing-away $p=1.00$, parallel $p=0.25$, perpendicular $p=1.00$; vision+proj facing $p=1.00$, facing-away $p=0.58$, parallel $p=1.00$, perpendicular $p=1.00$. Rows 15–18 are agreement-based: the committed two-stage pipeline scored agreement with the control’s verdicts (not accuracy against ground truth), so these rows are descriptive and are excluded from the confirmatory set and from the Holm family. SITE transfer tests (3: all/primary/secondary, §E) form a separate small family.

Table 6. Scaling (2B zs → 7B zs) and adaptation (7B zs → 7B General LoRA) family gains and bootstrap contrasts (pp, 95% CI).

Contrast vs. orientation	Point	95% CI	P(> 0)
Scaling (2B zs → 7B zs)			
horizontal	+13.9	[+2.9, +24.8]	0.993
depth	+5.5	[−5.7, +16.5]	0.834
containment	+5.1	[−5.9, +16.2]	0.820
topology/contact	−0.9	[−11.2, +9.4]	0.429
Adaptation (7B zs → 7B Gen. LoRA)			
depth	+5.0	[−6.5, +16.5]	0.802
topology/contact	+2.0	[−9.6, +13.3]	0.634
containment	+1.5	[−10.1, +13.0]	0.594
horizontal	+0.5	[−10.7, +11.5]	0.538

consistency on a strict pair means opposite verdicts.

Wilson CIs. All per-relation and SITE subset accuracies are reported with Wilson 95% confidence inter-

vals (main text and committed JSONs).

5. SITE Protocol, Keyword Rule, and Distributions

Pre-specification. Protocol frozen 2026-08-09 before any evaluation; example IDs committed in results/site/site_protocol.json; the protocol configuration is recorded in results/site/run_metadata.json. Evaluation-only: no SITE data used for training. Evaluation: Qwen2-VL-7B-Instruct frozen, greedy, official multiple-choice prompt and letter parser, max_new_tokens=16, images resized to ≤392px long side (constant protocol parameter; the installed transformers build ignores max_pixels), SDPA attention.

Subsets. Primary = official category spatial relationship reasoning (1,721 examples; 993 images). Secondary = heuristic orientation keywords (3,272; 2,024

Table 7. Facing-pair verdict patterns by condition ($n=103$); “exp.” = expected under verdict independence given the observed base rates.

Condition (facing pairs, $n=103$)	both-T	both-F	comp.	orig-T	flip-T	exp. both-F	exp. comp.
7B zero-shot	0	65 (63.1%)	38 (36.9%)	0.20	0.17	66.5%	30.2%
MiMo-V2.5 zero-shot ($n=100$)	0	56 (56.0%)	44 (44.0%)	0.30	0.14	60.2%	35.6%
LM-only LoRA	25 (24.3%)	10 (9.7%)	68 (66.0%)	0.62	0.52	18.0%	49.4%
HardNeg LoRA	18 (17.5%)	5 (4.9%)	80 (77.7%)	0.60	0.52	19.0%	50.2%
Projector LoRA	18 (17.5%)	15 (14.6%)	70 (68.0%)	0.53	0.50	23.5%	50.3%
Vision+Proj LoRA	14 (13.6%)	23 (22.3%)	66 (64.1%)	0.49	0.43	27.1%	49.4%

images). Exploratory = official movement prediction & navigation (1,103; 248 images). Overlaps (image): primary $\cap$ secondary = 488, secondary $\cap$ exploratory = 186, primary $\cap$ exploratory = 0; union = 2,591 images, each evaluated once.

Exact orientation-keyword rule (applied, case-insensitive, word- boundary regex, to question text plus non-image option text; src/datasets/site.py):

orientation, oriented, orient,
facing, face, faces, direction,
directional, view, viewpoint,
view angle, angle, turned, rotate,
rotated, rotation, toward, towards,
away from, left, right, front,
behind, in front of, parallel,
perpendicular, clockwise,
counterclockwise

Secondary subset composition ($n=2,024$ images) by official category: multi-view & cross-image reasoning 866 (raw 37.8%, CAA 8.8%); spatial relationship reasoning 488 (71.3%, 53.5%); object localization & positioning 346 (46.5%, 28.3%); movement prediction & navigation 186 (49.5%, 24.3%); 3D information understanding 91 (27.5%, 0.4%); counting & existence 47 (53.2%, 37.1%). Top sources: exoego4d 316 (36.4%), egoexo4d 288 (34.7%), MMTBench 237 (48.9%), SAT 183 (46.4%), MMIU 133 (28.6%), CLEVR 106 (92.5%), SPEC 102 (50.0%), SpatialEval 100 (53.0%), ThreeDSRBench 95 (61.1%), MMERealWorld 89 (32.6%), SeedBench 63 (58.7%), GQA 41 (75.6%), VStarBench 37, Blink 35, plus 27 smaller sources.

Decision rule (pre-specified). (1) If the primary subset shows the same broad weakness: run the VSR-trained 7B General LoRA on SITE (pre-specified step 2). (2) Else if only the orientation subset shows the effect: narrow the claim to object/direction orientation. (3) Else: report benchmark-specificity. Outcome: branch 2; the claim was narrowed accordingly. The VSR-trained 7B General LoRA adapter was subsequently evaluated on the identical SITE examples as a post-protocol cross-dataset

transfer check (no SITE training; predictions in results/site/vsr_lora_predictions.csv, paired statistics from scripts/compare_site_zeroshot_lora.py, canonical table in results/tables/site_validation_table.md):

Entries are raw accuracy / CAA (%). The adapter is statistically neutral overall, does not improve the orientation subset, and significantly degrades the official spatial-relationship category — VSR gains transfer poorly and can induce negative transfer.

Composition-confound analysis (CPU-only, zero-shot predictions). Because the orientation slice is keyword-derived and heterogeneous, we test whether the heuristic flag predicts lower correctness after controlling for task composition (logistic regression over the 2,591 zero-shot predictions; scripts/site_confound_analysis.py $\rightarrow$ results/site/orientation_confound_analysis.json):

Within the official spatial relationship reasoning category, orientation-flagged questions remain descriptively 7.5pp harder than unflagged ones (71.3% vs. 78.8%; $n=488/505$). No source dataset has $n \geq 30$ in both arms (orientation-heavy corpora are almost entirely flagged, others almost never), so source indicators absorb most of the orientation variance in the full model. Reading: the adjusted association is in the expected direction but not significant; the low aggregate heuristic score is largely explained by task composition. We therefore do not treat SITE as independent confirmation of the VSR orientation bottleneck, and the main text reports the slice as exploratory.

High-precision post-hoc subset (exploratory only). Restricting the heuristic to VSR-construct terms only (facing, facing away, parallel, perpendicular; word-boundary match on question or option text) yields $n=177$ examples (77 in the official spatial-relationship category, 99 in multi-view/cross-image reasoning, 1 in object localization): raw accuracy 44.1% (95% CI [37.0, 51.4]), CAA -2.3% . This subset is explicitly post-hoc and does not replace the pre-specified frozen definition; it is reported as exploratory only.

Table 8. SITE VSR-LoRA transfer check (post-protocol; predictions in results/site/vsr_lora_predictions.csv, canonical table in results/tables/site_validation_table.md). Subsets are defined by the frozen protocol IDs (see note below); no keyword reconstruction is used in analysis.

Subset	Zero-shot	VSR-LoRA	Δ	McNemar p
All images ($n=2,591$)	54.2 / 31.1	53.8 / 30.5	-0.4	0.66
Primary: spatial-rel. ($n=993$)	75.1 / 59.2	71.6 / 53.4	-3.5	0.004
Secondary: orient. kw. ($n=2,024$)	48.3 / 24.0	48.7 / 24.5	+0.4	0.70
Exploratory: movement ($n=248$)	48.8 / 23.1	50.4 / 25.6	+1.6	0.64

Table 9. Composition-confound logistic regressions (2,591 zero-shot SITE predictions; outcome = correct).

Model: correct $\sim$ orient + controls	OR (orient)	95% CI	p
Unadjusted	0.31	[0.25, 0.38]	2.4×10^{-28}
+ official category	0.74	[0.57, 0.96]	0.023
+ category, source, modality, $n_{options}$	0.84	[0.60, 1.16]	0.29

6. Failure Taxonomy

The 48 persistent-failure statements are the union of: wrong in both 7B zero-shot and 7B LoRA (20); wrong in ≥ 3 of 4 base conditions (38); wrong in both 2B and 7B zero-shot (28). The taxonomy below was produced by an automated audit (LLM-generated labels; commit 4b83729); its image grounding is unverified, and its reliability is assessed against two independent human re-audits below. The automated audit suggests many persistent errors remain visually interpretable while also identifying viewpoint and annotation ambiguities; it is qualitative evidence, not a causal localization of the failure. Annotations committed in results/failure_annotations.csv and results/orientation_persistent_annotations.csv.

Failure mode	n	Share
Clear image, model reasoning failure	18	37.5%
Camera viewpoint ambiguity	8	16.7%
Parallel/perpendicular geometry	6	12.5%
Annotation questionable (label wrong)	5	10.4%
Intrinsic orientation ambiguous	4	8.3%
Front/back object ambiguous	4	8.3%
Small / occluded object	2	4.2%
Subject/reference inversion	1	2.1%

Table 10. Persistent-failure taxonomy, 48 annotated cases.

Transition counts. Scaling (2B $\rightarrow$ 7B zero-shot): 28 both wrong, 23 2B-wrong/7B-right, 22 2B-right/7B-wrong, 64 both right. Adaptation (7B zero $\rightarrow$ 7B LoRA): 20 both wrong, 30 fixed, 27 broken, 60 both right. Of the 48 annotated cases, hard-negative training fixes 5; 15 still fail under all three 7B conditions. Facing/facing-away account for 68.8% of persistent failures.

Human re-audit and inter-source agreement (ad-

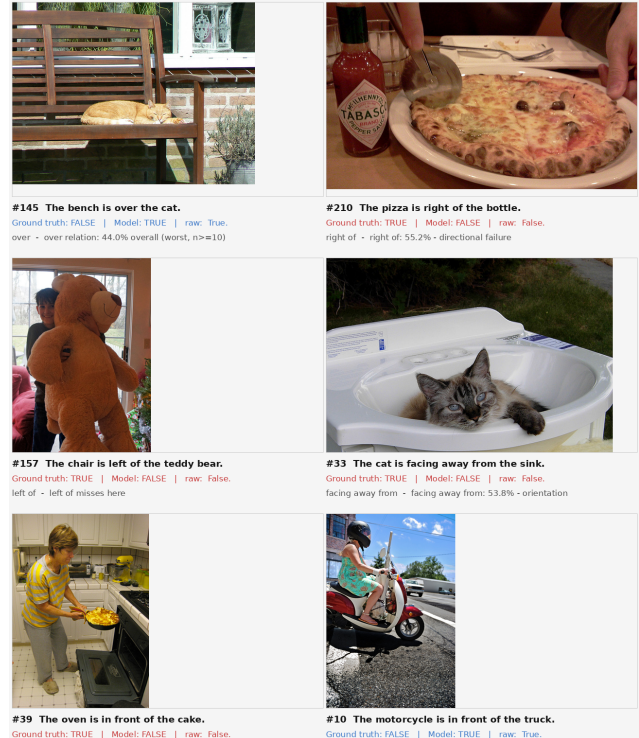


Figure 1. Representative persistent failures (annotated grid; 15 cases). Images: VSR test set. Per the automated audit (corroborated by the human re-audit below), the most frequent class is clear-image reasoning failure: the pose is visually interpretable to a human and the model still errs.

divitive reliability evidence). Because the clean-label audit (§3) and the taxonomy above originate from an automated (LLM-generated) audit, two independent human annotators re-rated the same examples from the images under a blind protocol (rater slots in scripts/iaa_tool.py): scripts/export_iaa_sheets.py

433 exports two rating sheets (the 137-example ori- 486
434 entation test set for the binary clean/ambiguous 487
435 flag, and the 48 persistent-failure cases for the 488
436 eight-class taxonomy) containing only image, state- 489
437 ment, and relation—ground truth, model predictions, 490
438 and the automated audit’s labels are withheld—and 491
439 scripts/iaa_tool.py provides a blind annotation in- 492
440 terface with auto-save. Agreement is computed by 493
441 scripts/compute_iaa.py (Cohen’s kappa for the binary 494
442 flag, Krippendorff’s alpha, nominal, for the taxon- 495
443 omy, both with bootstrap 95% confidence intervals; re- 496
444 sults/iaa/iaa_results.json).

445 Human–human agreement. The two human 498
446 annotators agree substantially: on the full 137- 499
447 example binary flag, 82.5% agreement ($\kappa=0.65$, CI 500
448 [0.51,0.77]); on the 48-case audited subset, 70.8% 501
449 agreement ($\kappa=0.44$, CI [0.21,0.67]). On the eight- 502
450 class taxonomy the exact-class agreement is 85.4% 503
451 (Krippendorff’s $\alpha=0.81$, CI [0.68,0.92]; commit- 504
452 ted at results/iaa/rater2_rater3_*). The binary 505
453 clean/ambiguous judgment is therefore the fuzzier of 506
454 the two tasks, while the failure taxonomy is reliably 507
455 rateable by humans. 508

456 Results (human vs. automated audit). On the 509
457 48 cases rated by both sources (the persistent-failure 510
458 set; the remaining 89 orientation examples were never 511
459 flagged by the automated audit and are excluded from 512
460 the agreement statistics), the binary clean/ambiguous 513
461 flag agrees on 32/48 cases (66.7%; $\kappa=0.36$, bootstrap 514
462 95% CI [0.13,0.59]). On the eight-class taxonomy the 515
463 exact-class agreement is 15/48 (31.2%; $\alpha=0.14$, CI 516
464 [−0.03,0.28]). The disagreement is systematic rather 517
465 than random: the human annotators never used sub- 518
466 ject/reference inversion (one used camera-viewpoint 519
467 ambiguity only three times), over-used annotation 520
468 questionable (11 and 9 vs. 5) and small/occluded object 521
469 (11 and 13 vs. 2), and flagged 62 and 60 of the 137 ori- 522
470 entation examples as ambiguous, versus 30 excluded by 523
471 the automated audit’s strict set. Part of the disagree-
472 ment on annotation questionable is attributable to the
473 blind protocol itself: judging whether a ground-truth
474 label is wrong requires the label, which was withheld.

475 Consequences. The frozen automated-audit table 524
476 (Table 2) is retained unchanged; we do not redefine 525
477 the exclusion sets to manufacture agreement. Both 526
478 human re-audits reproduce the qualitative conclu- 527
479 sions with their own taxonomy-derived strict masks 528
480 (Table 3, human A: best 77.6%, $n=107$; Table 4, 529
481 human B: best 78.8%, $n=104$; both leave 2B struc- 530
482 tured prompting without gain at 51.4%/51.0%), 531
483 so the sensitivity result is robust at the aggre- 532
484 gate level across two independent humans despite 533
485 the automated audit’s low per-item agreement 534

with either of them. An annotator-dependence 486
analysis (results/iaa/consensus_subsets.json, 487
scripts/iaa_consensus_subsets.py) shows the same 488
pattern: under the humans’ binary masks the clean- 489
subset gains largely vanish (best 66.7% human A / 490
66.2% human B vs. the 66.4% full-set best), while 491
under the consensus union mask ($n=62$) the trained 492
conditions again improve (7B General LoRA 77.4%, 493
7B Hard-Neg 74.2%, 2B zero-shot 72.6%) but 7B 494
zero-shot does not (58.1%). The statement that 2B 495
structured prompting shows no clean-subset gain 496
(53.3% on both full and strict sets) is specific to the 497
taxonomy-derived masks: under the human binary 498
consensus mask it reaches 69.3%. Overall the human 499
re-audits provide partial but aggregate-level corrobo- 500
ration, so the clean-label analysis should be read as 501
annotator-specific exploratory evidence rather than a 502
ruling on label noise. No reported accuracy number is 503
altered by this protocol. 504

LLM audit reliability note. We also ran the audit 505
prompts through a second LLM (MiMo-V2.5, image- 506
grounded; results/mimo/audit_mimo_*.csv). Its la- 507
bels collapsed to a single dominant class (43/48 508
taxonomy cases and 132/137 binary flags), yield- 509
ing no agreement with either the automated first- 510
pass labels or either human re-audit ($\kappa=0.00$; 511
 $\alpha \approx -0.1$; results/iaa/human_mimo_results.json, re- 512
sults/iaa/llm_mimo_results.json). We report this as 513
evidence that naive LLM audit prompting can degen- 514
erate to the first option, and we do not substitute it for 515
the human re-audit. 516

Released resource. The layered multi-rater data 517
— the automated audit, two independent human re- 518
audits, and the second-LLM audit-reliability check, 519
together with all per-item agreement statistics — is 520
versioned and released as a citable artifact under re- 521
sults/iaa/ (MANIFEST.json, ratings CSVs, agreement 522
JSONs, and the blind-rating tooling). 523

524 7. Reproducibility and Artifact Index

Code: anonymized archive accompanying this submis- 525
sion (no external links). Every headline number in this 526
paper maps to a committed artifact; the audit script 527
scripts/make_paper_figures.py recomputes the main 528
tables directly from the raw prediction CSVs (output 529
archived at paper/fig/audit_output.txt). 530

Environments. VSR/7B and SITE runs: Ubuntu 531
24.04, Python 3.12.13, torch 2.12.1+cu130, trans- 532
formers 5.14.1, datasets 5.0.1, peft 0.20.0, 1× RTX 533
A6000 (48 GB). 2B runs: earlier python3.10-era 534
environment (SmolVLM2 wrapper). MiMo-V2.5: 535
public API (OpenCode Go gateway), thinking 536
disabled, 12 parallel workers, per-request token ac- 537

Headline number	Artifact
74.0 / 76.6 / 76.5 / 68.3 (2B)	smolvlm2_*_metrics_*.json
80.9 / 84.7 / 84.3 / 82.9 / 83.1 (7B)	*_metrics_*.json
62.8 / 63.5 / 65.7 / 66.4 (orient. family)	same + prediction CSVs
Per-relation cells (Table 1 (main))	orientation_analysis.json
48-failure taxonomy	orientation_persistent_*.json
Probe tables (App. 3)	probe/*.json
Two-stage (App. D, agreement-based rows 15–18)	two_stage_results.json
Consistency (Fig. 4 (main))	consistency_stats_all.json
Vision-side p-values	vision_side_comparison.json
SITE numbers	site/zeroshot_7b_predictions.csv
	site/zeroshot_image_metrics.json
MiMo-V2.5 VSR / consistency / audit (Table 1 row, verdict table, App. 3)	mimo/*.csv, mimo/mimo_vsr_summary.json
Human re-audits + inter-source agreement (App. 6)	iaa/rater2_*.csv, iaa/rater3_*.csv, iaa/iaa_results.json, iaa/consensus
Bootstrap family contrasts (§D)	bootstrap_family_contrast.json
MiMo format sensitivity (§A)	mimo/mimo_parseable_sensitivity.json

Table 11. Headline-number → artifact → commit index (all paths relative to results/).

538 counting in results/mimo/usage_report.json (total
539 spend reported there). No lock files exist; re-
540 quirements.txt at the repository root declares the
541 pinned top-level dependencies. Seeds recoverable
542 from the archive: canonical VSR split seed 42
543 (dataset.train_test_split calls in the pipelines), IAA
544 bootstrap seed 20260811 (scripts/compute_iaa.py),
545 family-contrast bootstrap seed 20260812
546 (scripts/bootstrap_family_contrast.py), two-
547 stage CV random_state=42. Training-run
548 seeds were not recorded in the metrics JSONs
549 (a known limitation); the prepared multi-
550 seed extension (scripts/run_seed_variance.py /
551 scripts/aggregate_seed_variance.py) records and
552 reports per-run seeds explicitly.

553 Canonical freeze. The experimental result set
554 is a frozen snapshot. Canonical tables are gener-
555 ated by scripts/build_canonical_tables.py; clean-
556 label results by scripts/clean_label_orientation.py
557 (automated-audit table) and
558 scripts/build_human_clean_label_table.py (ver-
559 sioned human re-audit table); publication fig-
560 ures by scripts/make_publication_figures.py;
561 SITE paired transfer statistics by
562 scripts/compare_site_zeroshot_lora.py;
563 VSR/SITE headline recomputation by
564 scripts/make_paper_figures.py (output archived
565 at paper/fig/audit_output.txt); MiMo-V2.5 evalua-
566 tion and analysis by scripts/eval_mimo_vsr.py and
567 scripts/mimo_analysis.py (predictions and summaries
568 under results/mimo/).

569 Two-stage reproducibility caveat. The two-
570 stage pipeline requires per-patch embeddings
571 (results/probe/patch_embeddings.pkl, 8.2 GB, not
572 committed). Regeneration command (GPU, ≥40 GB):
573 python scripts/extract_patch_embeddings.py

574 from the repository root, then python
575 scripts/two_stage_reasoning.py; a SHA-256 checksum
576 of the generated file will be recorded in the release
577 (results/probe/patch_embeddings.pkl.sha256) once
578 it is regenerated. The committed evidence is the
579 aggregate results/two_stage_results.json (agreement-
580 based, see App. D); scripts/two_stage_reasoning.py
581 additionally scores per-example ground-truth
582 accuracy and saves per-example decisions
583 (results/two_stage_predictions.csv) when rerun,
584 so the ground-truth variant is produced by the same
585 regeneration step.

586 SITE secondary-subset audit note (corrected
587 before submission). An earlier derived analysis
588 (compare_site_zeroshot_lora.py v1, which fed
589 the initial zeroshot_image_metrics.json-adjacent
590 figures and the first canonical site table) re-
591 constructed the secondary subset by applying
592 ORIENTATION_KEYWORDS to the question text
593 only, yielding $n=1,824$ (47.3% raw / 22.6% CAA).
594 The frozen protocol’s pre-specified definition — re-
595 sults/site/site_protocol.json → frozen_ids.secondary
596 (3,272 examples; 2,024 with images), intersected
597 with the 2,591 evaluated IDs — is the authoritative
598 subset, and all numbers in this paper use it ($n=2,024$;
599 zero-shot 48.3% / 24.0% CAA; VSR-LoRA 48.7% /
600 24.5%, $p=0.70$). The derived question-only figure was
601 corrected before submission; both definitions support
602 the same conclusion (orientation $\ll$ primary), but only
603 the frozen-ID set matches the pre-specification. No
604 prediction CSV was modified and no model was rerun.

605 Data licenses/access. VSR: cam-
606 bridgelt/vsr_random (HF). SITE: franky-
607 veteran/SITE-Bench, CC-BY-4.0 (media in 130 GB
608 zips; requires HF read token). LoRA adapters are
609 committed and loadable with peft; base-model weights

610 via HF Hub.
611 Exact artifacts for the probe tables
612 (App. 3): results/probe/probe_results.json,
613 results/probe/patch_probe_results.json, re-
614 sults/probe/grounded_probe_results.json, re-
615 sults/probe/clear_subset_results.json, re-
616 sults/probe/grounded_boxes.json.
617 Runbook (one documented command per head-
618 line result). From the repository root, with the
619 requirements.txt environment: canonical VSR ta-
620 bles python scripts/build_canonical_tables.py;
621 headline recomputation + archive python
622 scripts/make_paper_figures.py > pa-
623 per/fig/audit_output.txt; automated clean-label
624 table python scripts/clean_label_orientation.py;
625 human re-audit tables python
626 scripts/build_human_clean_label_table.py -
627 rater rater2 and -rater rater3; inter-source
628 agreement python scripts/compute_iaa.py (+
629 scripts/iaa_consensus_subsets.py); consistency
630 tables python scripts/consistency_verdict_table.py
631 and scripts/eval_consistency_flips.py; family con-
632 trasts python scripts/bootstrap_family_contrast.py;
633 MiMo sensitivity python
634 scripts/mimo_parseable_sensitivity.py;
635 SITE transfer statistics python
636 scripts/compare_site_zeroshot_lora.py; two-stage
637 (requires the regenerated embeddings) python
638 scripts/two_stage_reasoning.py. A CI-style smoke
639 check runs the headline audit on a fresh clone; all
640 prediction CSVs are committed, so the numerical
641 results reproduce without any model or GPU.